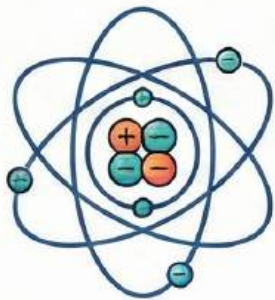


A Visual Guide to Electric Charge & Fields

@ThunderStudy



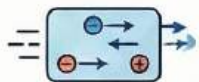
An intrinsic property from electron transfer, exerting electrostatic forces.



Quantization: $Q = ne$

$$e = 1.6 \times 10^{-19} \text{ C}$$

(Total charge =, total charge as integer multiple)



Conservation

Charge can neither be created nor destroyed.

Fundamental Interactions

Like charges repel, opposite charges attract.

The value of an electric charge does not change with the speed of the body.

1. The Basics of Electric Charge

Electron Transfer



Three Methods of Charging

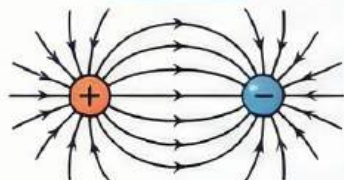


Friction

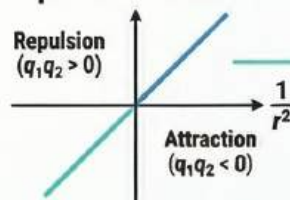
Conduction

Induction

Invariance



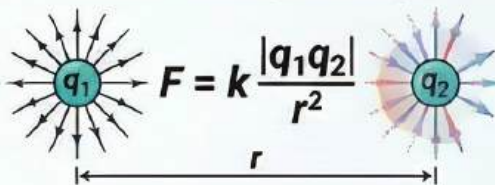
Graph of Force vs. $1/r^2$



Larger $|q_1q_2|$ = Steeper slope

2. Coulomb's Law: The Force Between Charges

Quantifies electrostatic force between stationary point charges.



$$F = k \frac{|q_1q_2|}{r^2}$$

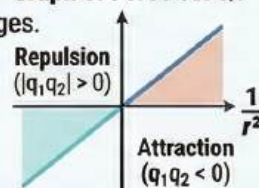
Quantifies electrostatic force between stationary point charges.

$$k = 9 \times 10^9 \text{ Nm}^2/\text{C}^2$$

Limitations

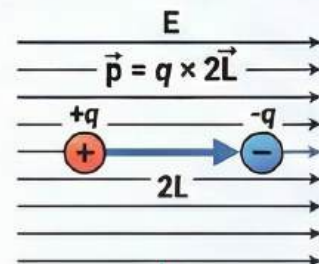
- ✓ Point charges
- ✓ Charges at rest
- ✓ Distance > nuclear size

Graph of Force vs. $1/r^2$



Larger $|q_1q_2|$ = Steeper slope

4. Electric Dipoles



Torque on a Dipole:

$$\tau = pE \sin \theta$$

Torque is maximum at 90° and zero at 0° & 180° .

Field on Axial Point:

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{2p}{r^3}$$

(stronger, same direction as p)

Field on Equatorial Point:

$$E = \frac{1}{4\pi\epsilon_0} \cdot \frac{p}{r^3}$$

(weaker, opposite direction to p)

Potential Energy:

$$U = -pE \cos \theta$$

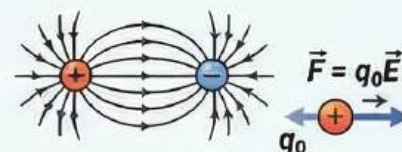
● Min ($-pE$, 0° , stable)

● Max ($+pE$, 180° , unstable)

3. The Electric Field & Field Lines

What is an Electric Field?

The region around a charge where another charge experiences an electrostatic force.



Torque on a Dipole:

$$\tau = pE \sin \theta$$

Turning force, max at 90° , zero at 0° & 180°

Properties of Electric Field Lines

- ✓ Imaginary
- ✓ Start +ve, end -ve
- ✓ Never intersect
- ✓ No closed loops
- ✓ Perpendicular to conductor surface

Potential Energy:

$$U = -pE \cos \theta$$

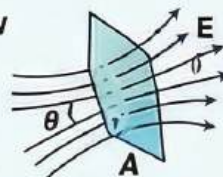
Min ($-pE$, 0° , stable) Max ($+pE$, 180° , unstable)

5. Electric Flux & Gauss's Law

What is Electric Flux (Φ)?

Total electric field lines passing perpendicularly through a surface.

$$\Phi = E \cdot A = EA \cos \theta$$

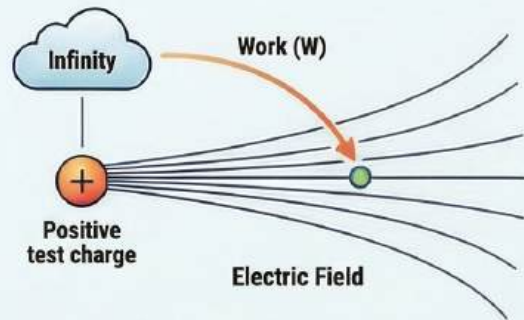


$$\text{Gauss's Law: } \Phi_{\text{closed}} = \frac{q_{\text{enclosed}}}{\epsilon_0}$$

Charge Distribution	Formula for Electric Field (E)	Key Characteristic
Infinitely Long Wire	$\frac{\lambda}{2\pi\epsilon_0 r}$	E is inversely proportional to distance (r).
Infinite Plane Sheet	$\frac{\sigma}{2\epsilon_0}$	E is constant and independent of distance.
Charged Spherical Shell	Outside: $\frac{kQ}{r^2}$ On Surface: $\frac{kQ}{R^2}$ Inside: 0	Field is zero inside, maximum on surface, decreases outside.

Mastering Electric Potential & Capacitance: A Visual Revision Guide

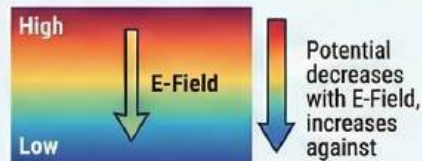
Fundamentals of Electric Potential (V)



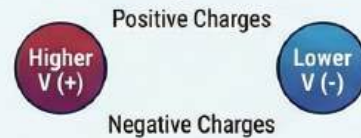
$$V = \frac{W}{q_0}$$

1 Volt
1 Joule of work per 1 Coulomb

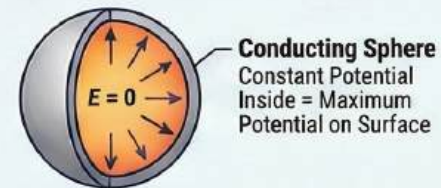
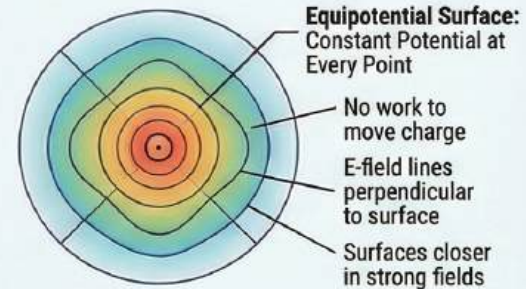
Potential and Electric Field Direction



Potential Difference Determines Charge Flow



Equipotential Surfaces & Conductors



Made By Dr Pragya

Capacitors and Capacitance

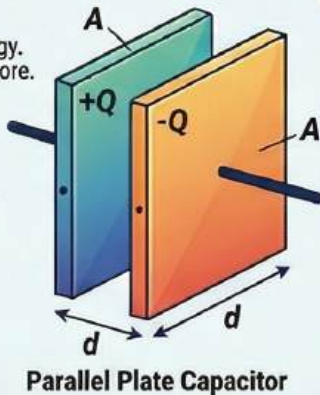
What is a Capacitor?

Device to store electrical energy.
Capacitance (C) is ability to store.

$$C = \frac{Q}{V} \text{ Farad (F)}$$

Depends on Geometry (A, d) and Medium
Not on Q or V

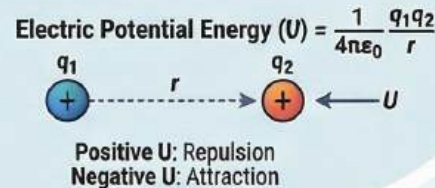
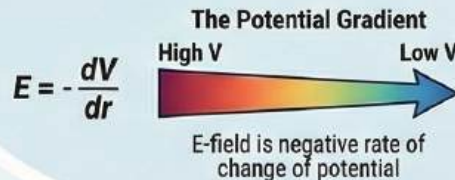
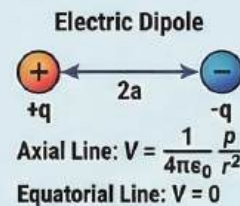
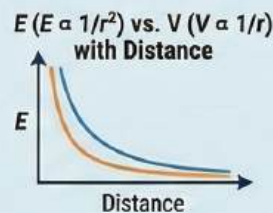
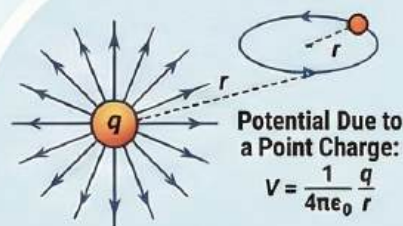
$$C = \frac{\epsilon_0 A}{d}$$



Energy Stored in a Capacitor

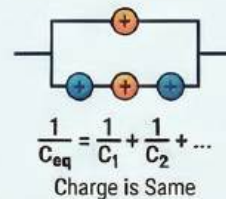
$$U = \frac{1}{2} CV^2 = \frac{Q^2}{2C}$$

Potential Calculations & Key Formulas

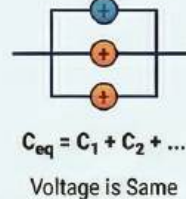


Combining Capacitors & The Effect of Dielectrics

Series Combination



Parallel Combination

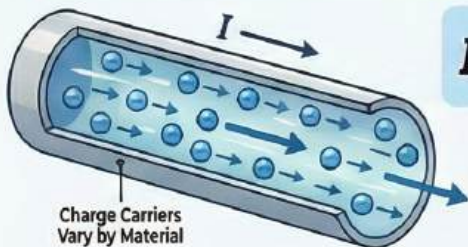


	Battery is Disconnected	Battery Remains Connected
Charge (Q)	Stays Constant (Q_0)	Increases by K ($K \cdot Q_0$)
Electric Field (E)	Decreases by K ($\frac{E_0}{K}$)	Stays Constant (E_0)
Potential (V)	Decreases by K ($\frac{V_0}{K}$)	Stays Constant (V_0)
Capacitance (C)	Increases by K ($K \cdot C_0$)	Increases by K ($K \cdot C_0$)
Energy (U)	Decreases by K ($\frac{U_0}{K}$)	Increases by K ($K \cdot U_0$)

@ThunderStudy

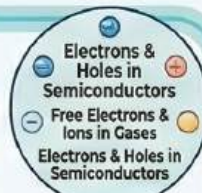
Current Electricity: A Visual Revision

Fundamentals of Electric Current



$$I = \frac{q}{t}$$

What is Electric Current?
Rate of flow of electric charge through a cross-section of a conductor.
Unit: Ampere (A) = Coulomb/second (C/s)



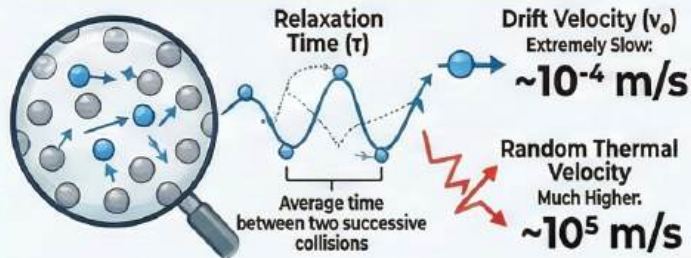
Current is a Scalar Quantity

Adds algebraically, not vectorially.
Direction is direction of positive charge flow.

Current Density (J)

$$J = \frac{I}{A}$$

The Microscopic View of Conduction



Higher Temperature = Shorter Relaxation Time (τ)

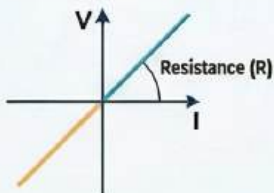
↑ Temp → ↑ Kinetic Energy
→ ↑ Collisions →
↓ Time between collisions.

Mobility (μ)
Measure of how quickly an electron can move.

$$\mu = \frac{v_d}{E}$$

Ohm's Law, Resistance & Resistivity

Ohm's Law
Current (I) is directly proportional to Potential Difference (V) (provided physical conditions constant).



$$V = IR$$

Resistance (R)
Obstruction to current flow from collisions.
Unit: Ohm (Ω)

Stretching vs. Adding Length

Adding Length: Length doubled = Resistance doubled

Stretching: Length doubled = Resistance becomes FOUR times larger (n^2 times)

Factors Affecting Resistance

$$R = \rho \left(\frac{L}{A} \right)$$

→ Directly proportional to Length (L)
→ Inversely proportional to Cross-sectional Area (A)

Resistivity (ρ)
Resistivity is a property of the material itself.

How Temperature Affects Resistance

Conductors (Metals)

↑ Temp →
→ ↑ Kinetic Energy →
→ ↑ Collisions →
→ ↓ Relaxation Time (τ)
→ ↑ Resistance.

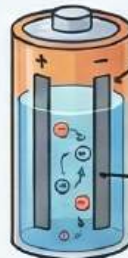
Alloys

Resistance increases slightly with temperature, but resistivity is very high.

Semiconductors & Electrolytes

↑ Temp →
→ Decreases resistivity and resistance
→ Better conductivity.

Cells: EMF & Internal Resistance



Electromotive Force (EMF or E)
Max potential difference when no current is drawn (open circuit)

Internal Resistance (r)
Resistance offered by electrolyte and electrodes to ion flow.

Factors Affecting Internal Resistance (r)

- ↑ Electrolyte Concentration
- ↑ Electrode Separation
- ↓ Temperature
- ↓ Larger Immersed Electrode Area

Discharging vs. Charging a Cell

Discharging: EMF > Terminal Voltage (V)
Charging: Terminal Voltage (V) > EMF

Complete Circuit Current

$$I = \frac{E}{R + r}$$

Essential Circuit Analysis Tools

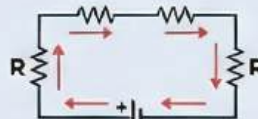
Kirchhoff's First Law (Junction Rule)



$$\sum I = 0$$

Algebraic sum of currents entering a junction is zero.
Based on conservation of charge.

Kirchhoff's Second Law (Loop Rule)



$$\sum \Delta V = 0$$

Algebraic sum of changes in potential around any closed circuit loop is zero.
Based on conservation of energy.

Wheatstone Bridge

A circuit with four resistors used to measure an unknown resistance accurately.

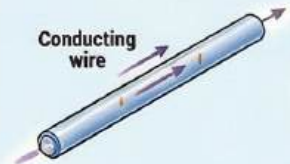


Balanced Condition

The bridge is balanced when no current flows through the central galvanometer.

Moving Charges & Magnetism: A Visual Revision

FUNDAMENTAL CONCEPTS

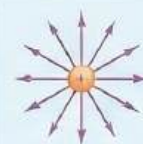


Electric Current Creates a Magnetic Field

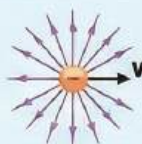
Oersted discovery: current-carrying conductor is surrounded by a magnetic field. (Magnetic effect of current)



Right-Hand Thumb Rule



Stationary Charge: Electric Field Only

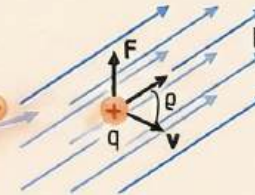
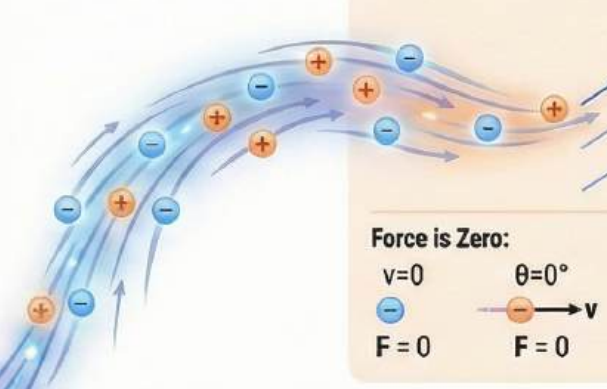


Moving Charge: Electric & Magnetic Fields



Current Conductor: Magnetic Field Only (No Net Charge)

FORCE ON A MOVING CHARGE (LORENTZ FORCE)



$$F = qvB\sin\theta$$

(or $F = q(\mathbf{v} \times \mathbf{B})$)

Force is Zero:

$$v=0$$

$$F=0$$

$$\theta=0^\circ$$

$$F=0$$

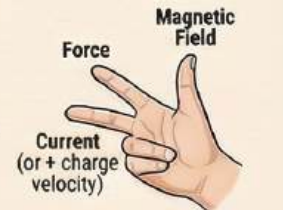
$$\theta=180^\circ$$

$$F=0$$

Maximum Force Condition:

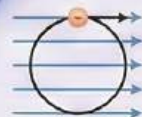
$$\theta=90^\circ$$

$$F = qvB$$



Fleming's Left-Hand Rule

MOTION OF A CHARGE IN A MAGNETIC FIELD

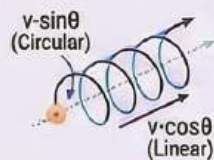


Path 1: Perpendicular Entry ($\theta=90^\circ$)

Follows a circular path. Speed & KE constant, velocity changes.

$$\text{Radius } (r) = \frac{mv}{qB}$$

$$\text{Time Period } (T) = \frac{2\pi m}{qB}$$



Path 2: Entry at an Angle (θ)

Helical path. $v\sin\theta$ causes circular motion, $v\cos\theta$ causes linear motion.

$$\text{Radius } (r) = \frac{mv\sin\theta}{qB}$$

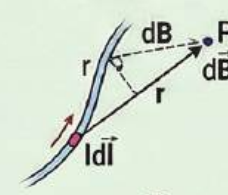
$$\text{Time Period } (T) = \frac{2\pi m}{qB}$$

(Independent of velocity!)

$$\text{Pitch } (P) = (v\cos\theta) \cdot T$$

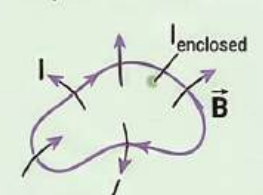
CALCULATING MAGNETIC FIELDS: KEY LAWS

Biot-Savart Law



$$dB = \frac{\mu_0}{4\pi} \frac{Idl \sin\theta}{r^2}$$

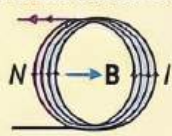
Ampere's Circuital Law



$$\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$

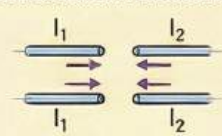
APPLICATIONS & DEVICES

Field at the Center of a Coil



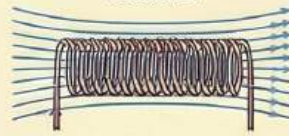
$$B = \frac{\mu_0 NI}{2r}$$

Force Between Parallel Wires



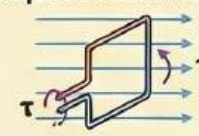
$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}$$

Solenoid



$$B = \mu_0 nI, \quad n = \frac{\text{turns}}{\text{length}}$$

Torque on a Current Loop

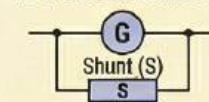


$$\tau = NIAB\sin\theta$$

($M=NI A$, magnetic moment)

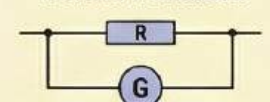
CONVERTING A GALVANOMETER

To create an **Ammeter**



Connect low resistance (S) in parallel to divert most current.

To create a **Voltmeter**



Connect high resistance (R) in series to limit current flow.

Magnetism & Matter: A Visual Revision Guide

1. The Fundamentals of a Magnet



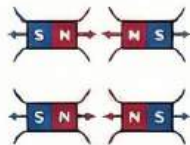
What is a Magnet?

Any substance that can create its own magnetic field and exert a force on other magnetic materials like iron and steel.

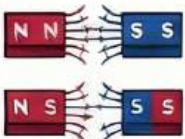
Directive Property



Pole Interaction



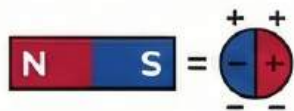
Pole Interaction



Monopoles Do Not Exist



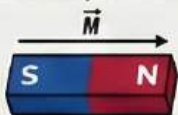
2. The Bar Magnet as a Dipole



An Analogy to the Electric Dipole

North Pole is analogous to a positive charge, and the South Pole to a negative charge.

Magnetic Dipole Moment



$$M = m \times 2l$$

(where 'm' is pole strength and '2l' is magnetic length).

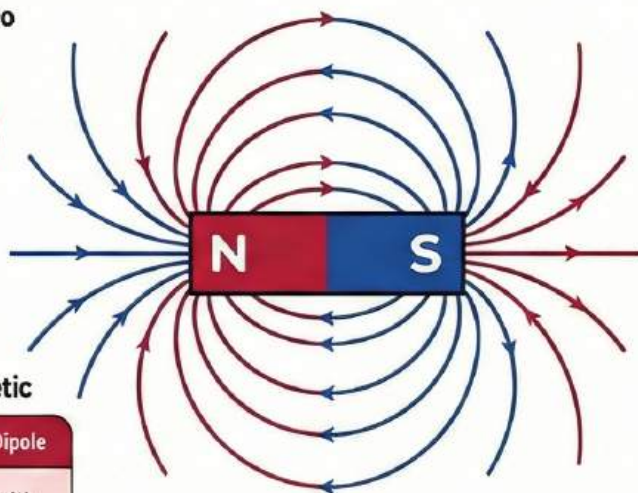
Key Formulas: Electric vs. Magnetic

Concept	Electric Dipole	Magnetic Dipole
Torque	$\tau = pE \sin(\theta)$	$\tau = MB \sin(\theta)$
Potential Energy	$U = -pE \cos(\theta)$	$U = -MB \cos(\theta)$
Work Done (Rotation)	$W = pE(\cos\theta_1 - \cos\theta_2)$	$W = MB(\cos\theta_1 - \cos\theta_2)$
Axial Field	$E = k \frac{2p}{r^3}$	$B = \frac{\mu_0}{4\pi} \frac{2M}{r^3}$
Equatorial Field	$E = k \frac{p}{r^3}$	$B = \frac{\mu_0}{4\pi} \frac{M}{r^3}$

3. Magnetic Fields & Gauss's Law

Magnetic Field Lines Form Closed Loops

They travel from N to S outside and S to N inside, completing a full loop. They never terminate.



Field Lines Never Intersect



If they did, it would mean two possible directions for the magnetic force at a single point, which is impossible.

Gauss's Law for Magnetism



Gauss's Law for Magnetism

The net magnetic flux through any closed surface is always zero. For every field line exiting, another must enter.

4. How Materials Respond to Magnetism



Intensity of Magnetization (I or M)

Net magnetic dipole moment developed per unit volume of a material in a magnetic field.

Unit: A/m.



Magnetizing Intensity (H)

Strength of external magnetic field's ability to magnetize a material.

Unit: A/m.



Magnetic Susceptibility (χ)

Measure of how easily a material can be magnetized.

$$\chi = I / H$$

Unitless.



Relative Permeability (μ_r)

Compares material's permeability to free space. Indicates ability to concentrate field lines.

$$\mu_r = 1 + \chi$$

Unitless.

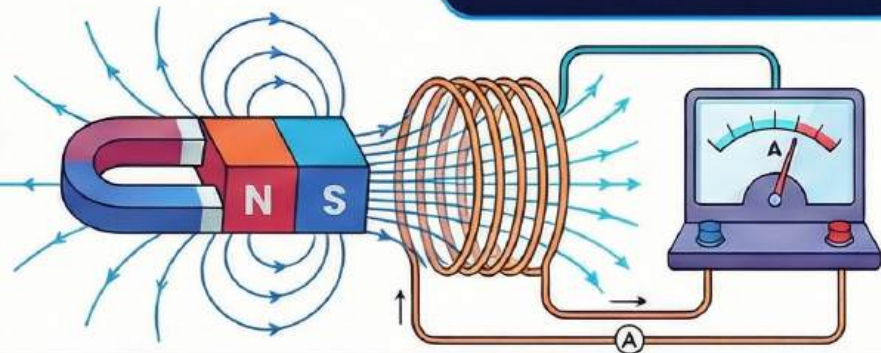
5. Classification of Magnetic Materials

How Field Lines Pass Through Materials



Property	Diamagnetic	Paramagnetic	Ferromagnetic
Interaction	Weakly repelled	Weakly attracted	Strongly attracted
Examples	Copper, Gold, Bismuth	Aluminum, Calcium, Oxygen	Iron, Nickel, Cobalt, Steel
Susceptibility (χ)	Small & Negative	Small & Positive	Large & Positive (>1)
Rel. Permeability (μ_r)	Slightly less than 1 ($0 < \mu_r < 1$)	Slightly greater than 1	Much greater than 1 ($\mu_r \gg 1$)
Effect of Temperature	Independent	Inversely proportional	Decreases, vanishes above Curie Temperature
Motion in Field	Moves from stronger to weaker parts	Moves from weaker to stronger parts	Moves rapidly from weaker to stronger parts

A Quick Guide to Electromagnetic Induction



THE FOUNDATIONS OF EMI

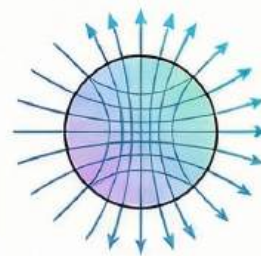
What is Electromagnetic Induction?

The process of producing an electric current (or EMF) in a coil by varying the magnetic field linked with it.



Discovered by Michael Faraday

Faraday demonstrated that moving a magnet near a coil, or vice-versa, caused a galvanometer to show a deflection, indicating a current.



@ThunderStudy

The Role of Magnetic Flux (Φ)

Magnetic flux is the measure of the total magnetic field lines passing through a given area. EMI occurs only when this flux changes.

Faraday's First Law



Whenever the magnetic flux linked with a coil changes, an EMF is induced in the coil. This EMF lasts as long as the change in flux continues.

Faraday's Second Law



The magnitude of the induced EMF is directly proportional to the rate of change of the magnetic flux. Faster change = more EMF.

FARADAY'S & LENZ'S LAWS

$$e = - \frac{d\Phi}{dt}$$

The induced EMF (e) equals the rate of change of magnetic flux ($d\Phi/dt$). The negative sign is explained by Lenz's Law.

Lenz's Law: The Law of Opposition

The direction of the induced current is always such that it opposes the very change in magnetic flux that produced it.

North Pole Approaching Coil



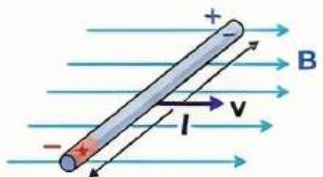
The coil's near face becomes a North pole to repel the magnet. This is achieved by an anti-clockwise current flow.

North Pole Moving Away



The coil's near face becomes a South pole to attract the magnet. This is achieved by a clockwise current flow.

Motional EMF

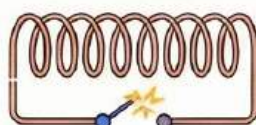


$$e = Bvl$$

B = magnetic field, v = velocity, l = length

KEY PHENOMENA & FORMULAS

Self-Induction



Self-Inductance of a Solenoid:

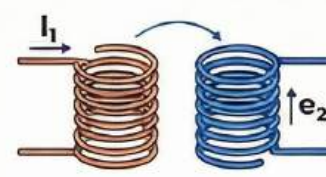
$$L = \mu_0 n^2 A l$$

Self-Inductance of a Solenoid

L = coefficient of self-inductance
 n = turn density
 A = area, l = length

This is why sparks can occur when a circuit is suddenly broken.

Mutual Induction



$$M = \mu_0 n_1 n_2 A l$$

The phenomenon where a changing current in one coil (primary) induces an EMF in a neighboring coil (secondary).

M

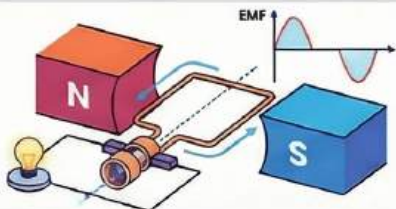
' M ' is the coefficient of mutual inductance, dependent on the properties of both coils.

AC Generator

A device that converts mechanical energy into alternating electrical energy based on the principle of electromagnetic induction.

How it Works

A coil is rotated within a uniform magnetic field. As it rotates, the magnetic flux through it changes continuously, inducing an alternating current.



APPLICATIONS OF EMI

Induced EMF in an AC Generator:

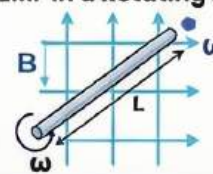
$$e = NBA\omega \sin(\omega t)$$

The EMF (e) varies sinusoidally with time.
The peak EMF is $e_0 = NBA\omega$.

Fleming's Right-Hand Rule



EMF in a Rotating Rod



Rotating Rod EMF:

$$e = \frac{1}{2} B\omega L^2$$

B = magnetic field
 ω = angular velocity
 L = length of the rod

ALTERNATING CURRENT: A VISUAL REVISION GUIDE

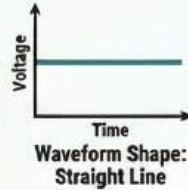
@ThunderStudy

AC vs. DC: THE FUNDAMENTALS

Direct Current (DC)

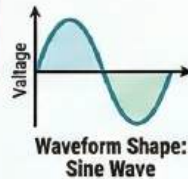
Flows in one direction.

In DC, the magnitude and direction of current are constant, meaning electrons flow steadily in a single direction.



Alternating Current (AC)

Periodically reverses direction. In AC, electrons oscillate back and forth. The magnitude changes continuously, while the direction reverses after a fixed time interval.



Frequency in India is 50 Hz for AC. This means one, the frequency of DC is always 0 Hz.

UNDERSTANDING AC VALUES: PEAK vs. RMS

I_0 (Peak Value)

RMS value is the 'effective' value of AC. The RMS (Root Mean Square) value of an AC current is the equivalent DC current that would produce the same amount of heat in the same resistor.

I_{rms} (Effective Value)

Household ratings are RMS values. If a device label says 10A AC, it refers to the RMS value, not the maximum (peak) current.

Relating RMS and Peak Values:

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

220V AC is more dangerous than 220V DC. A 220V DC shock is constant at 220V. For 220V AC (which is the RMS value), the peak voltage reaches $220V \times \sqrt{2} = 311V$, delivering a more severe shock.

KEY AVERAGES & DERIVATIONS

Average current over a full cycle is ZERO.

The positive half-cycle cancels out the negative half-cycle, resulting in a net average value of zero.

This is why AC doesn't produce a chemical effect.

Average current over a half cycle is not zero.

Formula: $I_{avg} = 0$

The total charge transferred over one complete cycle is zero.

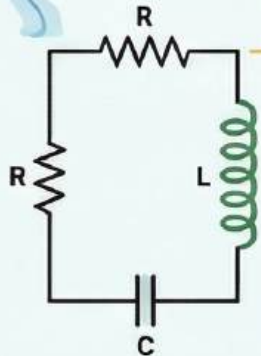
ESSENTIAL AC DEVICES & CIRCUITS

LCR CIRCUIT & IMPEDANCE

In a series circuit with a resistor (R), inductor (L), and capacitor (C), the total opposition to current is called impedance (Z).

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

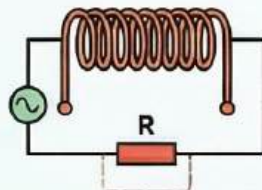
Electrical Resonance: Resonance occurs when the circuit's impedance is at its minimum ($Z=R$), allowing maximum current to flow. This happens when inductive reactance (X_L) equals capacitive reactance (X_C).



CHOKE COIL

Controls current efficiently.

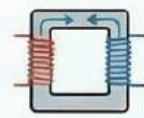
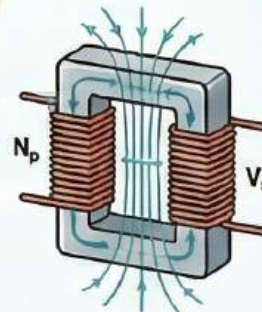
A choke coil is an inductor used to control AC with minimal power loss, as it opposes current via self-induction rather than dissipating energy as heat like a resistor.



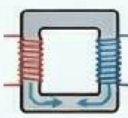
TRANSFORMER

Changes AC voltage levels.

Working on the principle of mutual induction, a transformer can increase (step-up) or decrease (step-down) AC voltage. It does not work with DC.



Step-Up: $N_s > N_p$
Voltage Increase



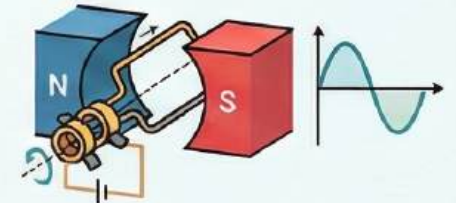
Step-Down: $N_s < N_p$
Voltage Decrease

Transformers conserve power. In an ideal transformer, input power equals output power ($V_p I_p = V_s I_s$).

AC GENERATOR

Converts mechanical to electrical energy.

Based on the principle of electromagnetic induction, it generates AC by rotating a coil in a magnetic field, which continuously changes the magnetic flux.

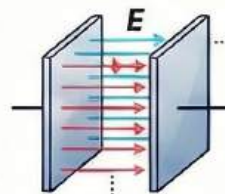


How it Works: As the coil rotates, the direction of the induced current (determined by Fleming's Right-Hand Rule) reverses every half rotation, creating alternating current.

A Visual Guide to Electromagnetic Waves

@ThunderStudy

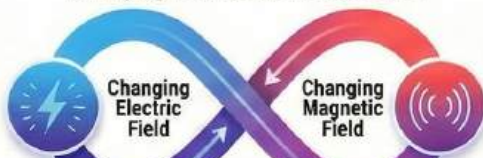
The Birth of an EM Wave



Changing Electric field Capacitor

The Missing Link: Displacement Current (Maxwell)
Discovered by Maxwell, this is the current produced by a changing electric field.

Changing Fields Create Each Other

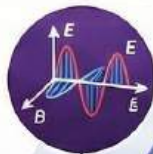


The Self-Perpetuating Cycle

A changing E-field creates a changing B-field, which in turn creates a new changing E-field. This cycle propagates through space, independent of the original source, turning on EM wave.

Core Properties of EM Waves

Transverse in Nature
The electric and magnetic fields oscillate perpendicular to each other and to the direction the wave is traveling.



No Medium Required
Unlike sound, EM waves can travel through the vacuum of space.



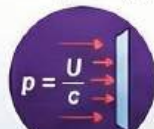
Travel at the Speed of Light
In a vacuum all EM waves travel at a constant speed, $c = 3 \times 10^8$ m/s.



Electrically Neutral
They have an energy and are therefore not deflected by electric or magnetic fields.



Carry Energy & Momentum
EM waves transport energy and exert a force, known as radiation pressure, on a surface.
Momentum (p) = Energy (U) / c



Maxwell's 4 Unifying Equations

1. Gauss's Law for Electricity



$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{enc}}{\epsilon_0}$$

Defines the net electric flux through a closed surface in the gaseous electric field, it confirms that single electric charges (monopoles) exist.

2. Gauss's Law for Magnetism



$$\oint \vec{B} \cdot d\vec{A} = 0$$

States that the net magnetic flux through any closed surface is zero. This implies that magnetic monopoles (isolated North or South poles) do not exist.

3. Faraday's Law of Induction



$$\oint \vec{E} \cdot d\vec{l} = -\frac{d\Phi_B}{dt}$$

A changing magnetic flux induces an electric field. This is the principle behind electric generators.

4. Ampere-Maxwell Law (Modified)



$$\oint \vec{B} \cdot d\vec{l} = \mu_0 (I_c + \epsilon_0 \frac{d\Phi_E}{dt})$$

A magnetic field is generated by both a current in a wire (conduction current) and by a changing electric field (displacement current).

Everyday Physics: EM Waves in Action



Why do welders wear special glasses?

To protect their eyes from the intense ultraviolet (UV) radiation produced by the welding arc, which can cause serious eye damage.



How do microwaves heat food?

The oven produces microwaves at the resonant frequency of water molecules. This causes the water in the food to vibrate rapidly, generating heat.



Why do radio signals bend around buildings but light doesn't?

Diffraction (bending) occurs when a wave's wavelength is comparable to the size of an obstacle. Radio waves have long wavelengths (meters) similar to buildings, while light's wavelength is microscopic.

The Electromagnetic Spectrum

The Inverse Relationship:

Highest Frequency / Shortest Wavelength

As you move from Gamma rays to Radio waves across the spectrum, frequency and energy decrease, while wavelength increases.

Lowest Frequency / Longest Wavelength



Gamma Rays

Source:

Source: Radioactive Decay

Uses: Cancer therapy, sterilizing medical equipment, detecting cracks in engines



X-Rays

Source:

Source: Fast-moving electrons hitting a metal target

Medical imaging (bone fractures), airport security scanners



Ultraviolet (UV)

Source:

Source: The Sun, special lamps

Sterilization, food preservation, detecting forged currency, treating skin conditions



Visible Light

Source:

Source: The Sun, light bulbs

Vision, photography, fiber optics



Infrared (IR)

Source:

Source: Hot bodies, molecular vibrations

Thermal imaging, remote controls, physical therapy (heat lamps), night vision



Microwaves

Source:

Source: Magnetron tubes

Microwave ovens, radar systems, telecommunications, Wi-Fi



Radio Waves

Source:

Source: Oscillating electronic circuits

Radio broadcasting, television signals, remote-controlled toys, MRI

Made By Dr Pragya

Ray Optics: A Visual Guide to Light

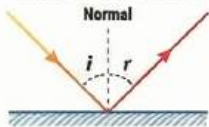
@ThunderStudy

1. Fundamentals of Light

What is Optics?

The branch of physics that studies the behavior and properties of light, and the construction of instruments that use or detect it.

Laws of Reflection

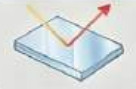


1. The angle of incidence (i) equals the angle of reflection (r).
2. The incident ray, reflected ray, and the normal all lie in the same plane.

Reflection vs. Refraction

Bouncing back of light from a surface

Bending of light as it passes from one medium to another



1. The incident ray, reflected ray, and the normal all lie in the same plane.
2. Snell's Law: The ratio $\sin(i)/\sin(r)$ is a constant, known as the refractive index (μ).

2. Mirrors & Images

Real vs. Virtual Images



Real images can be formed on a screen and are always inverted.



Virtual images cannot be formed on a screen and are always erect.

The Mirror Formula:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

Magnification (m):

$$m = \frac{\text{Height of Image}}{\text{Height of Object}} = -\frac{v}{u}$$

A negative ' m ' indicates a real, inverted image.

Mirror Types	Concave Mirror	Convex Mirror
Action on Light	Converges (focuses) light	Diverges (spreads) light
Image Type	Mostly Real & inverted (one case of Virtual & Erect)	Always Virtual, Erect & Diminished
Common Uses	Shaving mirrors, dentist's tools, headlights	Vehicle side-view mirrors, security mirrors
Focal Length	Negative	Positive

3. Refraction, Lenses & Total Internal Reflection (TIR)

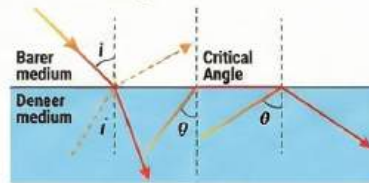
Refractive Index (μ)

A measure of how much light slows down in a medium.

$$\mu = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

Total Internal Reflection (TIR)

1. Light travels from a denser to a rarer medium.
2. Angle of incidence is greater than the critical angle.



Total Internal Reflection (TIR)

The Lens Formula:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

Lens Maker's Formula:

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

Power of a Lens (P):

The ability of a lens to converge or diverge light, measured in Diopters (D).

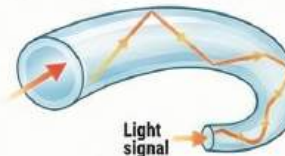
$P = 1/f$ (where f is in meters).

Positive for convex, negative for concave.

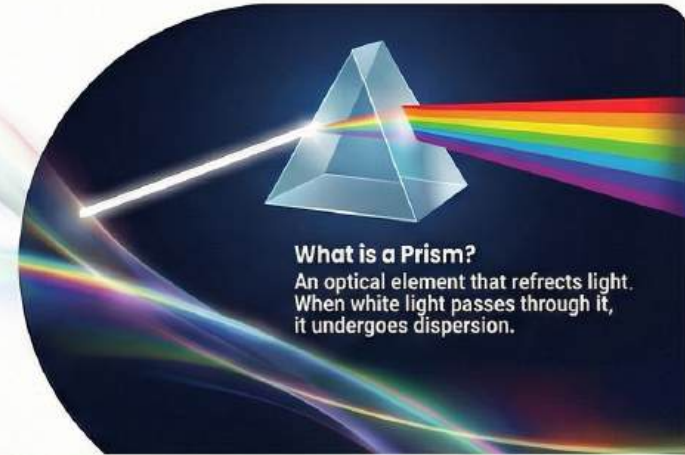
Lens Types	Convex Lens	Concave Lens
Action on Light	Converges (focuses) light	Diverges (spreads) light
Shape	Thicker in the middle, thinner at the edges	Thinner in the middle, thicker at the edges
Image Type	Mostly Real & Inverted (one case of Virtual & Erect)	Always Virtual, Erect & Diminished
Common Uses	Magnifying glasses, microscopes, cameras	Eyeglasses for nearsightedness, telescopes



Application of TIR: Optical Fibers
Used to transmit data over long distances as light signals, which undergo continuous TIR within a glass or plastic fiber.

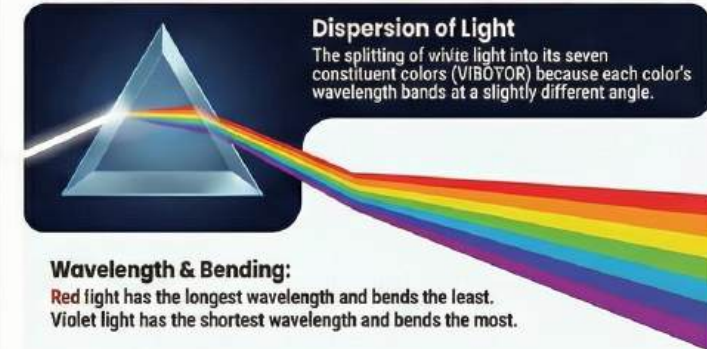


4. The Prism



What is a Prism?

An optical element that refracts light. When white light passes through it, it undergoes dispersion.



Dispersion of Light

The splitting of white light into its seven constituent colors (VIBGYOR) because each color's wavelength bends at a slightly different angle.

Wavelength & Bending:

Red light has the longest wavelength and bends the least.
Violet light has the shortest wavelength and bends the most.

The Prism Formula

Calculates the refractive index (μ) of the prism's material using the Angle of Prism (A) and the Angle of Minimum Deviation (δ_m):

$$\mu = \frac{\sin\left(\left(A + \frac{\delta_m}{2}\right)/2\right)}{\sin\left(\frac{A}{2}\right)}$$

TWO VIEWS OF LIGHT: THE BRANCHES OF OPTICS



RAY (GEOMETRICAL) OPTICS
Treats light as a straight ray.
Explains reflection, refraction.



WAVE OPTICS
Treats light as a wave. Explains interference, diffraction, polarization.

HISTORICAL THEORIES OF LIGHT



Newton's Corpuscular Theory
Proposed that light is composed of tiny, perfectly elastic particles called

SUCCESSSES & FAILURES

Success: Reflection, refraction.
Failure: Interference, diffraction, polarization.



Huygens' Wave Theory
Light is a wave propagating through an ether medium.

SUCCESSSES & FAILURES

Success: Reflection, refraction, interference, diffraction, polarization.
Failure: Photoelectric effect, light speed in denser media.

A Visual Guide to Wave Optics

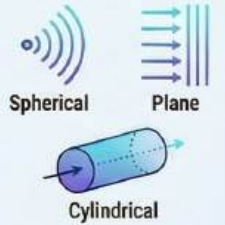
HUYGENS' PRINCIPLE: HOW LIGHT WAVES TRAVEL

What is a Wavefront?
Locus of all points oscillating in the same phase.

DIRECTION OF LIGHT

Step 1: Secondary Wavelets

3 Types of Wavefronts

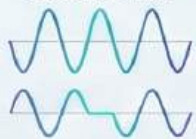


INTERFERENCE: WHEN LIGHT WAVES MEET

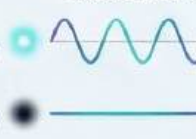
Superposition Principle: Resultant displacement is the vector sum of individual displacements.

What is Interference?
Modification in light energy distribution when coherent waves waves superpose.

CONSTRUCTIVE



DESTRUCTIVE

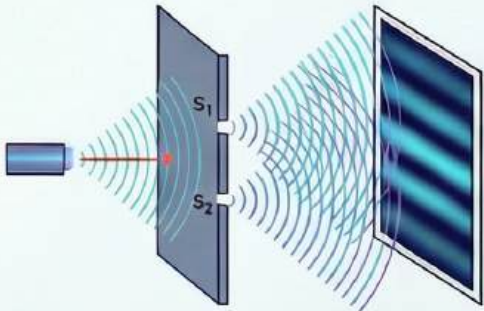


CONDITIONS FOR INTERFERENCE

Type of interference	Path Difference (Δs)	Phase Difference ($\Delta \phi$)
Constructive	$n\lambda$	$2n\pi$
Destructive	$(2n + 1)\lambda/2$	$(2n + 1)\pi$

($n = 0, 1, 2, 3, \dots$)

YOUNG'S DOUBLE SLIT EXPERIMENT (YDSE)



The First Proof of Light's Wave Nature

Alternating Bright and Dark Fringes
Predictable pattern of bright (constructive) and dark (destructive) bands.



Fringe Width (β)

$$\beta = \frac{\lambda D}{d}$$

- λ : wavelength
- D : screen-slit distance
- d : slit separation

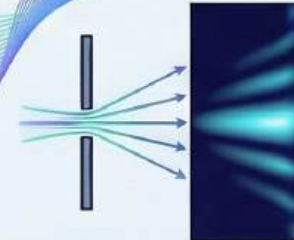
Energy is Redistributed, Not Created

The dark fringes are dark because their light energy has been redistributed to the bright fringes, making them extra bright. This adheres to the law of conservation of energy.

Coherent Sources are Essential

The experiment requires coherent sources, which emit light of the same frequency with a constant phase difference. This is achieved by splitting light from a single source into two.

DIFFRACTION: WHEN LIGHT BENDS



Bending Around Corners:
Light bends around obstacles, entering the shadow region.

When is Diffraction Noticeable?
Most prominent when obstacle size is comparable to wavelength (λ).

Diffraction Pattern vs. YDSE Pattern
Unlike uniform YDSE fringes, diffraction has a very bright, wide central maximum with much weaker secondary maxima.

THE DUAL NATURE OF MATTER & RADIATION: A VISUAL REVISION GUIDE

@ThunderStudy

The Core Concept: Light's Dual Nature

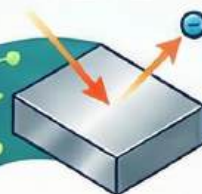


Wave Nature
Interference and diffraction demonstrate light's wave nature.

Light behaves as a wave in some phenomena and as a particle in others. This is called the Dual Nature of Radiation.

Particle Nature
The Photoelectric Effect demonstrates its particle nature.

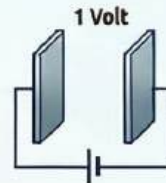
Understanding the Basics



Work Function (Φ_0 or W)
The minimum amount of energy required to eject a free electron from a metal surface.

Work Functions of Different Metals

- Cesium: 2.14 eV
- Potassium: 2.3 eV
- Platinum: 5.65 eV



What is an Electron Volt (eV)?
A small unit of energy, it's the energy an electron gains when accelerated by a potential difference of 1 Volt.

Energy Conversion
 $1 \text{ eV} = 1.6 \times 10^{-19} \text{ Joules}$

Four Ways to Eject an Electron (Electron Emission)



Thermionic Emission
Ejecting electrons by heating a metal (providing thermal energy).



Photoelectric Emission
Ejecting electrons (photoelectrons) by shining light of a suitable frequency on a metal.



Secondary Emission
A fast-moving "primary" electron strikes a surface, knocking out a "secondary" electron.



Field Emission
Pulling an electron out of a metal surface using a very strong electric field.

Einstein's Explanation: The Photon



Light is made of Photons
Einstein proposed that light consists of discrete energy packets called photons.

Photon Energy

$$E = h\nu$$

h = Planck's Constant ($6.67 \times 10^{-34} \text{ J}\cdot\text{s}$)

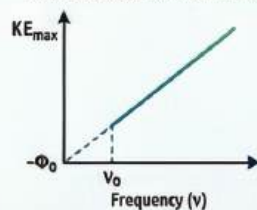
Properties of a Photon
Photons are chargeless, have zero rest mass, and travel at the speed of light.

Einstein's Photoelectric Equation

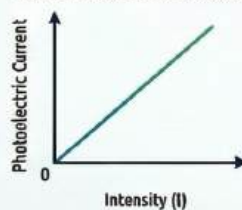
$$h\nu = \Phi_0 + KE_{\max}$$

Key Graphs of the Photoelectric Effect

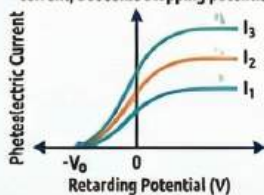
Kinetic Energy vs. Frequency
Straight line, starts at threshold frequency (ν_0), y-intercept is $-\Phi_0$.



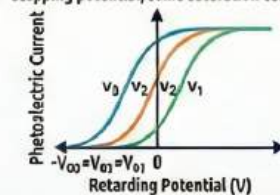
Current vs. Intensity
Straight line passing through origin; current is proportional to intensity.



Effect of Intensity on Stopping Potential
Higher intensity = higher saturation current, but same stopping potential.



Effect of Frequency on Stopping Potential
Higher frequency = more negative stopping potential, same saturation current.



De Broglie's Hypothesis: Matter has Waves Too

Every Moving Particle has a Wave
De Broglie proposed that any moving matter has an associated matter wave.



Why Don't We See Waves in Daily Life?

For large mass (m), wavelength (λ) is incredibly small and undetectable.

De Broglie Wavelength

$$\lambda = \frac{h}{p} = \frac{h}{mv}$$

Wavelength of an Accelerated Electron

$$\lambda = \frac{12.27}{\sqrt{V}} \text{ Angströms (\AA)}$$

For an electron accelerated by a potential difference V .

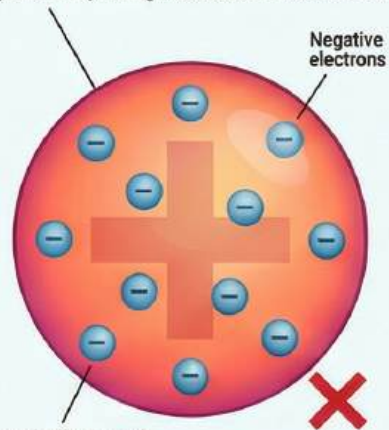
Made By Dr Pragya

The Evolution of the Atom: From Plum Pudding to Bohr's Orbits

@ThunderStudy

1. The Early Idea: Thomson's Model

The "Plum Pudding" Model: Proposed that an atom is a uniform sphere of positive charge with negatively charged electrons embedded throughout, similar to plums in a pudding .. or seeds in a watermelon.

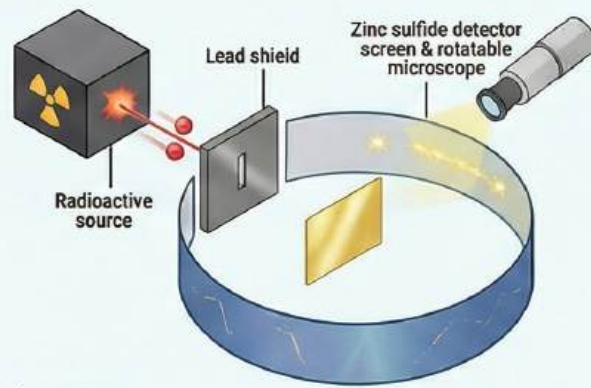


Electrically Neutral: The total positive charge of the sphere was balanced by the total negative charge of the electrons, resulting in a neutral atom.

Why It Failed: This model could not explain the results observed in Rutherford's subsequent gold foil scattering experiment.

2. The Game Changer: Rutherford's Gold Foil Experiment

The Experimental Setup: Alpha particles from a radioactive source were fired at a very thin gold foil, with a zinc sulfide screen and a rotatable microscope to detect where the particles landed.



Observation 1: Most Particles Passed Straight Through: The vast majority of alpha particles went through the foil with little to no deflection ($\pm 1^\circ$), suggesting the atom is mostly empty space.

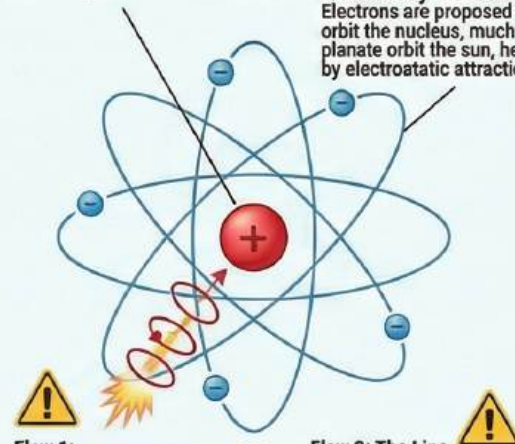
Observation 2: Some Particles Were Deflected: About 1 in 8000 particles were deflected by significant angles, indicating they encountered a repulsive force.

Observation 3: A Tiny Fraction Bounced Back: An extremely small number of particles (0.14% scattered $< 90^\circ$ with a mere few at 180°) were deflected back, suggesting a collision with something very dense and positively charged.

3. The Nuclear Model: Rutherford's Atom & Its Limitations

The Atom has a Dense, Positive Core: Rutherford concluded that all the positive charge and most of the atom's mass are concentrated in a tiny central region called the nucleus.

A "Planetary" Structure: Electrons are proposed to orbit the nucleus, much like planets orbit the sun, held by electrostatic attraction.



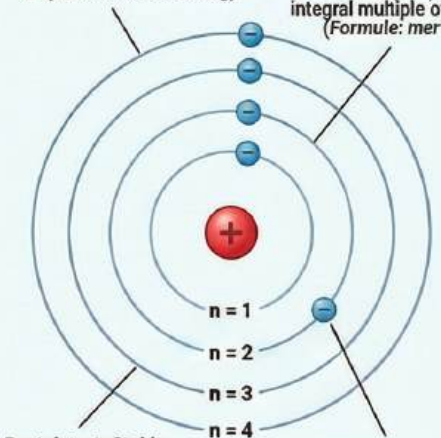
Flaw 1: The Instability Problem: According to classical physics, an orbiting electron should continuously radiate energy, lose speed, and spiral into the nucleus, causing the atom to collapse.

Flaw 2: The Line Spectroscopy Mystery: The model could not explain why atoms emit light only at specific, discrete wavelengths (line spectra) when heated.

4. The Quantum Leap: Bohr's Atomic Model

Postulate 1: Stable Orbits: Electrons revolve in specific, stable orbits called "stationary orbits" where they do not radiate energy.

Postulate 3: Quantized Angular Momentum: An electron can only be in an orbit where its angular momentum (mvr) is an integral multiple of $\hbar/2\pi$. (Formula: $mvr = n\hbar$)



Postulate 1: Stable Orbits: Electrons revolve in specific, stable orbits called "stationary orbits" where they do not radiate energy.

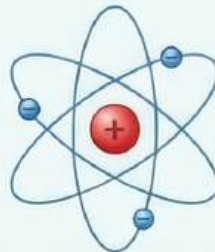
Postulate 3: Energy Transitions: An electron emits or absorbs energy (as a photon) only when it jumps between these allowed orbits. The photon's energy equals the energy difference between the orbits ($E = E_2 - E_1$).

5. Understanding Atomic Behavior: Key Concepts & Spectra

Key Concepts

Distance of Closest Approach: The minimum distance an alpha particle gets to the nucleus before its kinetic energy is fully converted to potential energy and it is repelled back. This helps estimate the size of the nucleus.

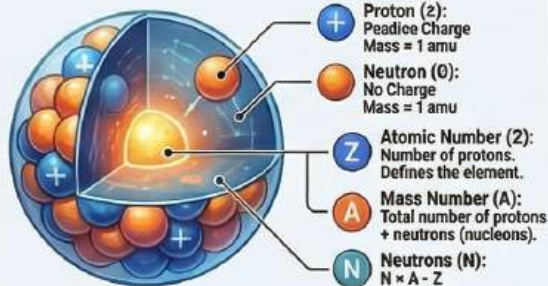
Impact Parameter: The perpendicular distance between the path of an alpha particle and the center of the nucleus. A smaller impact parameter results in a larger scattering angle.



Atomic Spectra: When atoms are excited, electrons jump to higher orbits and then fall back, emitting light of specific wavelengths. This creates a unique "line spectrum" for each element.

	Series Name	Final Orbit (n_1)	Originating Orbits (n_2)	Spectral Region
Lyman		1 (Innermost)	2, 3, 4... ∞	Ultraviolet
Balmer		2	3, 4, 5... ∞	Visible
Paschen		3	4, 5, 6... ∞	Infrared
Brackett		4	5, 6, 7... ∞	Infrared
Pfund		5 (Outermost)	6, 7, 8... ∞	Far Infrared

The Anatomy of a Nucleus



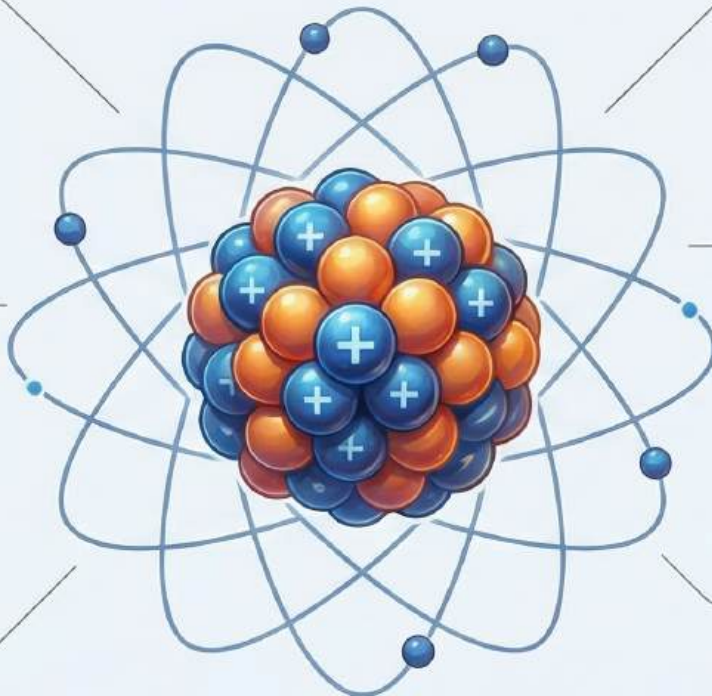
The Nucleus: The Atom's Core

Control part, concentrated protons & neutrons, containing all positive charge and almost all mass.

Decoding Uranium-235 (^{235}U)

$^{92}_{92}\text{U}$ Z = 92 (82 protons)
 $^{92}_{92}\text{U}$ A = 235 (Total nucleons)
 N = 235 - 92 = 143 neutrons.

A Quick Guide to the Atomic Nucleus



Key Properties & Measurements



Universal Nuclear Density:
 $\sim 2.29 \times 10^{17} \text{ kg/m}^3$
 (Constant for all atoms, independent of size).

Nuclear Radius (R) = $R_0 A^{1/3}$
 ($R_0 = 1.2 \text{ Fermi or } 1.2 \times 10^{-16} \text{ m}$).



Nuclear Charge:
 Total positive charge = $+Ze$.

Atomic Mass Unit (amu or u):
 Standard unit, 1/18th mass of Carbon-12
 (1 amu = $1.66 \times 10^{-27} \text{ kg}$).

The Force That Binds



The Strong Nuclear Force

Powerful attractive force holding nucleons together, overcoming electrostatic repulsion.

Strongest Force in Nature

Charge Independent
 (acts equally p-p, n-n, p-n)

Extremely Short Range
 (effective $\approx 10 \text{ Fermi}$, rapidly decreases & becomes repulsive).

Classifying Nuclei

Isotopes



Same protons (Z), different neutrons.

Isotones



Same neutrons, different protons.

Isobars



Same mass number (A), different protons & neutrons.

Mass, Energy, and Stability



Mass Defect (Δm)

Nucleus mass < Total mass of individual nucleons. "Missing" mass converts to energy.

$$E = mc^2$$

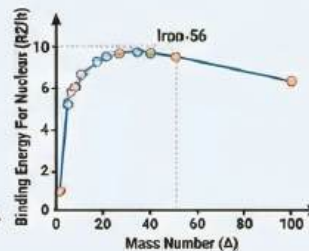
Binding Energy (BE):

Energy released when nucleus forms. Energy equivalent of mass defect.

The Magic Number:
 1 amu = 931 MeV BE.

Higher BE/A = More stable nucleus.

Iron-56 (Fe) is most stable ($\sim 0.6 \text{ MeV}$).



Nuclear Reactions: Fission vs. Fusion

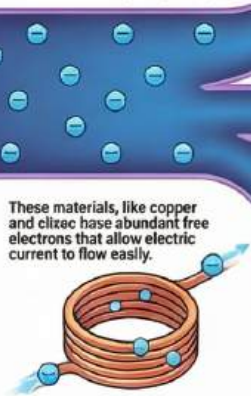


Process	Heavy nucleus splits into lighter ones	Two light nuclei combine to form heavier one
Trigger	Heavy nucleus absorbs slow neutron	Extremely high temperatures & pressure
Example	Uranium-235 splitting	Hydrogen isotopes fusing to Helium (Sun)
Energy Release	Huge (e.g., $\sim 200 \text{ MeV}$ per U-235)	Even more per nucleon than fission
Byproducts	More neutrons, can cause chain reaction	Neutrons or other particles
Application	Nuclear power plants, atomic bombs	Stars, experimental reactors (ITER)

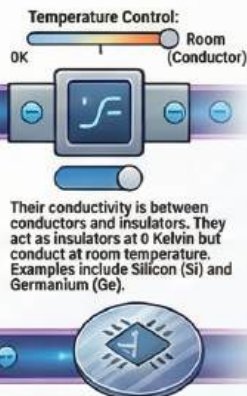
A Quick Guide to Semiconductors: From Silicon to Circuits

1. MATERIALS CLASSIFIED BY CONDUCTIVITY

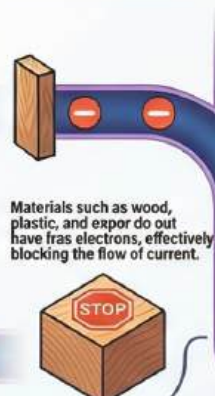
CONDUCTORS:
The Electron Superhighways



SEMICONDUCTORS:
The Smart Gates



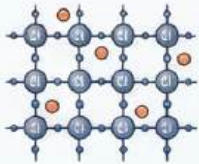
INSULATORS:
The Electrical Roadblocks



2. TYPES OF SEMICONDUCTORS: PURE VS. DOPED

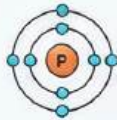
INTRINSIC (Pure) SEMICONDUCTORS

These are made of a single, pure element like Silicon. Current is generated when heat provides enough energy to break covalent bonds, creating an equal number of free electrons and "holes".

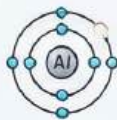


EXTRINSIC (Impure) SEMICONDUCTORS

N-TYPE: Creating Excess Electrons
Formed by adding pentavalent impurities (5 valence electrons), like Phosphorus, to Silicon. This creates a surplus of free electrons, which become the majority charge carriers.

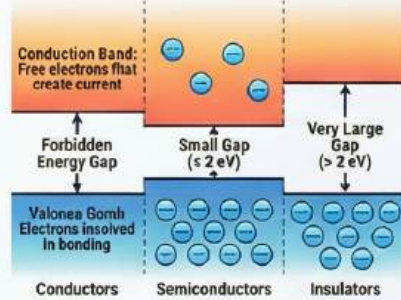


P-TYPE: Creating "Holes"
Formed by adding trivalent impurities (3 valence electrons), like Aluminum, to Silicon. This results in a deficit of electrons, creating positively charged "holes" which act as majority charge carriers.



3. THE ENERGY BAND THEORY

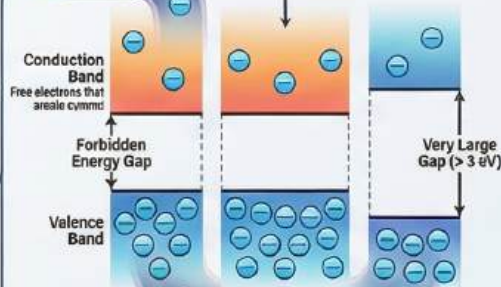
Understanding Electron Energy Levels:
Electrons in a crystal exist in energy ranges called bands.



Understanding Electron Energy Levels:
Electrons in a crystal exist in energy ranges called bands.

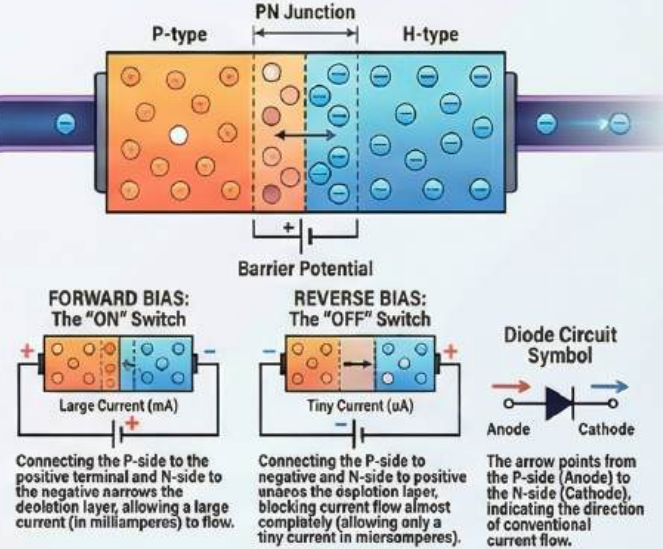
Energy Gaps Determine Conductivity:
The "Forbidden Energy Gap" between these bands dictates if a material conducts.

Conductors Zero Gap
Semiconductors Small Gap (< 3 eV)
Silicon (Si) 1.1 eV
Germanium (Ge) 0.7 eV
Insulators Very Large Gap (> 3 eV)



4. THE PN JUNCTION: HEART OF THE DIODE

Formation of the Depletion Layer:
When P-type and N-type materials are joined, electrons and holes diffuse across the junction, creating a central region with no mobile charge carriers called the depletion layer. This layer establishes a "Barrier Potential"



5. APPLICATION: RECTIFIERS - CONVERTING AC TO DC

What is a Rectifier? An electronic circuit that converts alternating current (AC) into one-directional Direct Current (DC), using the one-way current property of diodes.

HALF-WAVE RECTIFIER



Pulsating DC

Uses a single diode to allow only one half of the AC wave to pass through, blocking the other half. This results in a pulsating DC output with gaps.

FULL-WAVE RECTIFIER



Pulsating DC

Uses two or more diodes to convert both halves of the AC input wave into a continuous, pulsating DC output, making it more efficient.

-DC-

**Join thunder study official On
Telegram For More Study**

Click Here To Join